

Internet Protocols Explained

In the second part of this series of articles on internet protocols, we shall delve into the journey from unsecure protocols to secure protocols, with a focus on securing the transport and application layers in the OSI model.



The Transmission Control Protocol (TCP) is a transport protocol, as it dictates the way data is sent and received. Before transmitting data, TCP opens a connection with the recipient and ensures that all packets arrive in order once the transmission begins. It is via TCP that the recipient acknowledges receiving each packet that arrives. Missing packets are sent again if receipt is not acknowledged. Hence this is a reliable protocol, but not fast.

TCP and IP were originally designed to be used together, and these are often referred to as the TCP/IP suite. However, other transport protocols can be used with IP. The TCP header has a default size of 20 bytes. Up to 40 bytes of options can be added.

TCP flow

Step 1: Establish the connection

TCP needs a valid connection as a prerequisite. To establish a connection, it follows a three-way handshake method.

- First, the requesting client sends the server a SYN packet including SEQ.

- SYN: SYN stands for synchronise.
- SEQ: SEQ is a unique, random number that ensures full transmission in the correct order and without duplicates.
- Once the server has received the packet, it agrees to the connection by returning a SYN-ACK packet. The ACK includes the client's sequence number plus 1. The client will send some random SEQ number.
- ACK: ACK stands for acknowledging its own sequence number to the client.
- Finally, the client acknowledges the receipt by sending its own ACK packet, which in this case contains the server's sequence number plus 1. The client can now begin transferring data to the server.

Step 2: Transmit the packets

After the communication gets established, the TCP data is sent. Each packet transferred with data requires acknowledgement; hence the data and sequence are sent. The receiver sends the acknowledgement by increasing the acknowledgment number by the size of the packet received. The sequence and acknowledgement packet make sure the data is received without loss and prevent duplicates.

To avoid lost packets, the sender starts the timer after sending the packet and puts it in the retransmission queue. If the timer runs out and the sender doesn't receive the acknowledgement, the packet is resent. Here, even if the receiver gets multiple packets due to the slow speed, it will discard the duplicates.

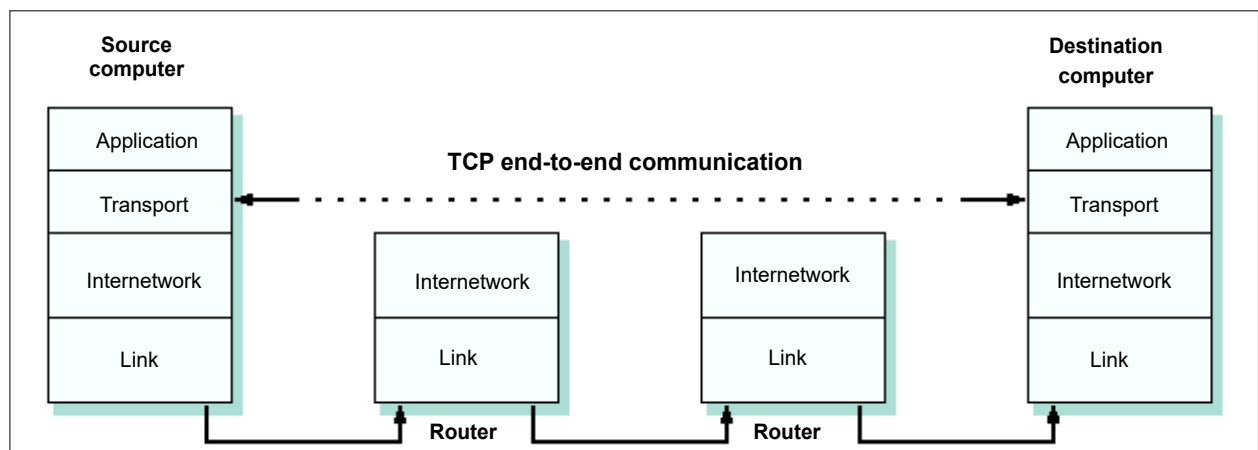


Figure 1: Transport layer (Credits: https://www.technologyuk.net/computing/computer-networks/internet/images/tcp_end_to_end_communication.gif)